

Grade 6 Accelerated
Unit 9
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

For questions 1-6, complete the table below.

	Data Description	Examples	Data Classification
1.	Scores on a Math test	70, 76, 78, 84, 92, 97	numerical
2.	Favorite classes taken by a student survey	Math, Science, Social Studies, Language Arts, Physical Education	categorical
3.	Shoe sizes of students (Answers may vary)	7.5, 8, 6, 6.5, 7.5, 8 (Answers may vary)	numerical
4.	Favorite football team (Answers may vary)	Steelers, Lions, Jaguars, Patriots (Answers may vary)	categorical
5.	Sports played after school	Football, cheerleading, soccer, baseball (Answers may vary)	categorical
6.	Number of students in each class period	18, 20, 22, 19, 18 (Answers may vary)	numerical

7. Categorical data is a type of data that is divided into groups. This type of data is qualitative.

For questions 8-10, classify each as Categorical or Numerical.

8. Number of teachers in each department at a large high school.

Numerical

9. Eye colors of all the students in your class.

Categorical

10. Height of your family members.

Numerical

For questions 1-7, determine whether each question is statistical or not statistical.

1. What Ocean does Florida border?

Not statistical

2. What size jerseys will each team at the school need?

Statistical

3. How many pets do you have?

Statistical

4. Is Orlando the capital of Florida?

Not statistical

5. How many followers do your classmates have on Twitter?

Statistical

6. What is each member of your class's favorite movie?

Statistical

(answers may vary)

7. The question "Who was the first President of the United States?" is not statistical. Use the definition to justify why it is not a statistical question.

The question cannot be answered by collecting data. Additionally, there is no variability in the data.

8. Decide if the question, "How many houses are in your neighborhood?" is statistical or non-statistical. Explain your decision.

Statistical. We could answer the question by collecting data and there would be variability in the data.

9. Create an example of a statistical question.

How many siblings do you have?

10. Create a counterexample that is NOT a statistical question.

What time does practice start?

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Lesson 3 Practice

Name Answer Key
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Disney World in Orlando Florida opened Magic Kingdom in 1971. Since then, it has become a world-renowned attraction. Use the table that shows the number of visitors recorded (in millions) for questions 1-3.

Year	Total Visitors (in millions)
2010	16.972
2012	17.536
2014	19.332
2016	20.395
2018	20.859
2020	6.941

$$\begin{array}{r}
 16.972 \\
 +17.536 \\
 \hline
 34.508 \\
 +19.332 \\
 \hline
 53.840 \\
 +20.395 \\
 \hline
 74.235 \\
 +20.859 \\
 \hline
 95.094 \\
 +6.941 \\
 \hline
 102.035
 \end{array}$$

$$\begin{array}{r}
 17.005 \\
 6 \overline{) 102.035} \\
 \underline{6} \\
 42 \\
 \underline{42} \\
 0035 \\
 \underline{0030} \\
 5
 \end{array}$$

- What is the mean attendance during this span of years? Round to the hundredths place.
17.01 total visitors (in millions)
- Interpret the mean attendance for the 6 years.
The average attendance was 17 million.
- Explain why the mean in the data above does not give information about Magic Kingdom's visitor attendance today.
The number of visitors in 2020 was much lower than normal, which pulled the mean down.

The 5 deepest lakes in the United States by depth are listed in the table below. Use the data for questions 4-5.

Lake	Total Depth (in feet)
Crater Lake	1,949
Lake Tahoe	1,645
Lake Chelan	1,486
Lake Superior	1,332
Lake Pend Oreille	1,150

$$\begin{array}{r}
 1949 \\
 +1645 \\
 \hline
 3594 \\
 +1486 \\
 \hline
 5080 \\
 +1332 \\
 \hline
 6412 \\
 +1150 \\
 \hline
 7562
 \end{array}$$

$$\begin{array}{r}
 1512.4 \\
 5 \overline{) 7562} \\
 \underline{5} \\
 25 \\
 \underline{25} \\
 06 \\
 \underline{5} \\
 12 \\
 \underline{10} \\
 20
 \end{array}$$

4. Find the mean of the data.
1512.40 total depth (in feet)
5. Interpret the mean depth of the lakes based on the data.
The average depth of the lake is 1512.40 ft.

The Tampa Bay Rays had 23 pitchers in their 2021 season. A random sample of 7 pitchers are listed in the table. Use the data for questions 6-7.

Pitchers Name	Pitchers Age
Corey Kluber	35
Luis Patino	22
Brooks Raley	33
Shane Baz	22
Nick Anderson	31
Ryan Yargrough	30
Josh Fleming	25

$$\begin{array}{r}
 28.28 \\
 7 \overline{) 198.0} \\
 \underline{14} \\
 58 \\
 \underline{56} \\
 20 \\
 \underline{14} \\
 60 \\
 \underline{56} \\
 4
 \end{array}$$

$$\begin{array}{r}
 57 \\
 + 33 \\
 \hline
 90 \\
 + 22 \\
 \hline
 112 \\
 + 31 \\
 \hline
 143 \\
 + 30 \\
 \hline
 173 \\
 + 25 \\
 \hline
 198
 \end{array}$$

6. Find the mean age of the pitchers given.
28 years
7. Interpret the mean based on the data given from the sample of pitchers.
The average age of pitchers is 28 years.

A farmer in northeast Florida raises chickens, pigs, and cows. He sells his chicken eggs to people in surrounding areas. He begins keeping a record of what each chicken lays per week to best estimate the amount he will have for sale from week to week. Use the data in the table for questions 8-10.

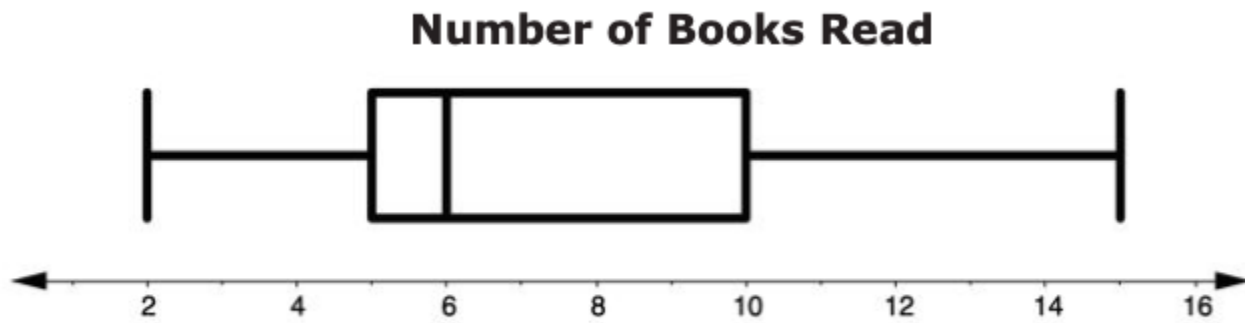
Eggs Laid by Chickens in One Week			
Chicken A:	6	Chicken E:	6
Chicken B:	5	Chicken F:	7
Chicken C:	9	Chicken G:	8
Chicken D:	3	Chicken H:	6

8. What number of eggs do most chickens lay at this farm?
6 eggs
9. What is the median amount of eggs produced?
6 eggs
10. What does the value of the median represent in this context?
Half of the chickens lay less than the median value of 6 eggs, and half of the chickens lay more than 6 eggs.

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Lesson 4 Practice

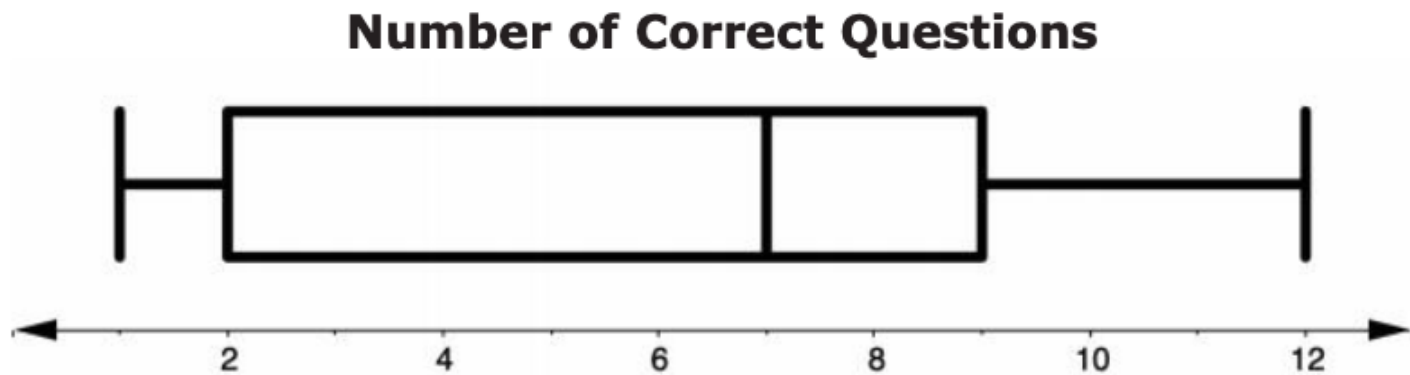
Name Answer Key
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A survey of a 6th grade class was given asking what number of books students read in the last two months of school. The data in the box plot shows the results. Use the box plot for questions 1-5.



1. The minimum books read is 2, while the maximum is 15.
2. The median number of books read is 6.
3. The upper quartile is 10.
4. The lower quartile is 5.
5. The range is $15 - 2 = 13$.
6. The Interquartile range (IQR) is $10 - 5 = 5$.
7. A box plot divides data into 4 equal parts that each contain 25% of data from a given data set.

A quiz was given to students in a science class. Use the data in the box plot for questions 8-10.

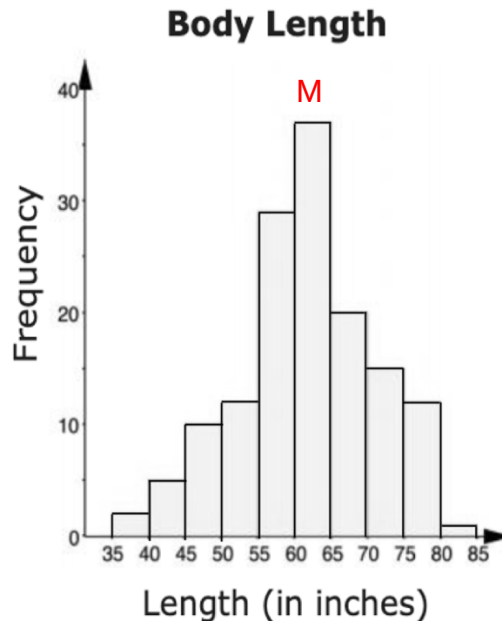


8. What percent of students got between 9 and 12 questions correct?
25%

9. What percent of students got between 2 and 9 questions correct?
50%

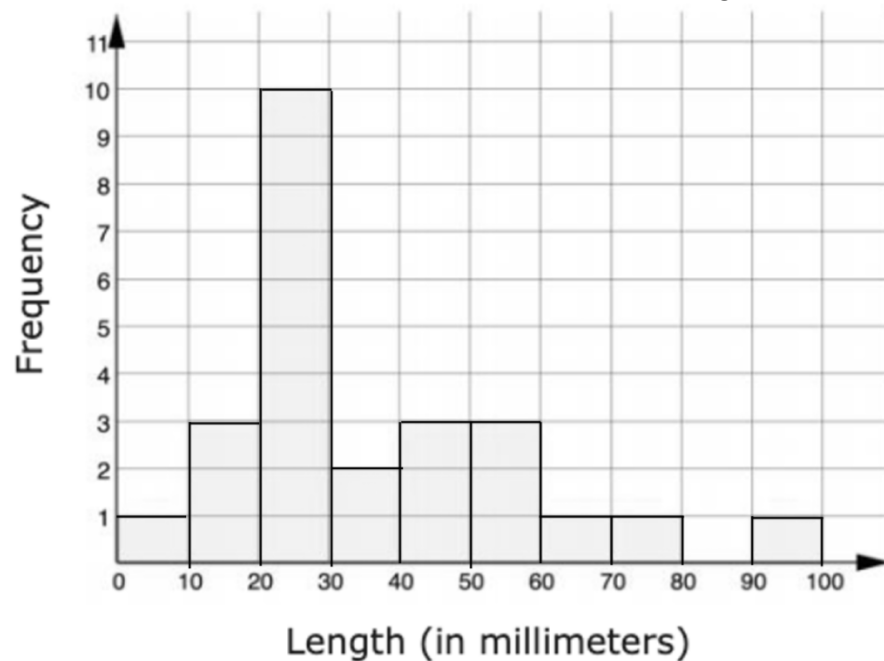
10. What percent of students got between 7 and 12 questions correct?
50%

In a study of wild bears, researchers measured the length, in inches (in.) of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- How many bears are between 45 and 50 inches long?
10 bears
- Which single range of length has the most bears?
60 – 65 inches
- What is the smallest possible length of a bear in this data set?
35 inches
- Are there more bears measuring less than 65 inches or more than 65 inches?
There are more bears measuring less than 65 inches.
- If bears older than 15 were added to the sample, hypothesize what would happen to the spread of the data.
I think the data would skew more to the right.
- Mark an “M” above the bar that contains the median.
Shown on histogram

An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of earthworms in one container. The farmer measured the earthworms in millimeters then created a histogram from the data.



7. Is it possible for the sample to contain an earthworm measuring more than 100 mm?
No
8. More than half of the earthworms measure less than 30 millimeters.
9. How many data pieces are in the histogram?
25 data points
10. What single range of lengths has the most earthworms?
20 – 30 millimeters

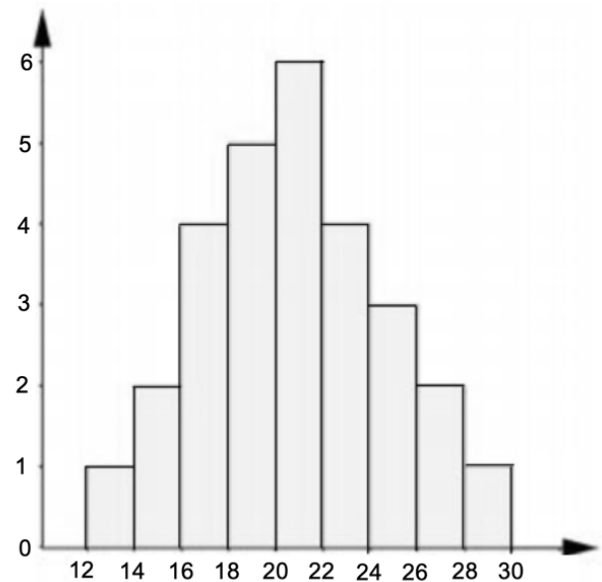
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Lesson 6 Practice

Name **Answer Key** _____
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1. A **right** skewed graph has a longer trail of data to the right and the peak of the data on the left.

2. The shape of the histogram is

- ☒ symmetric.
☐ left skewed.
☐ right skewed.

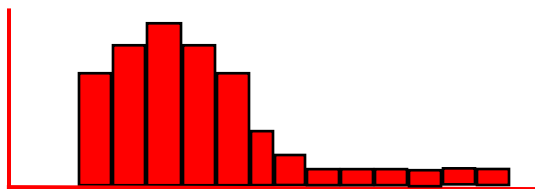


3. If the data value **197** were added to the histogram above, it would be considered an outlier.

Answers will vary

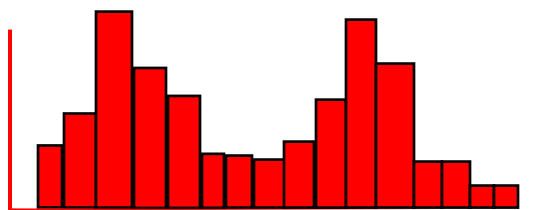
4. Sketch a histogram whose shape can be described as right skewed.

Answers will vary



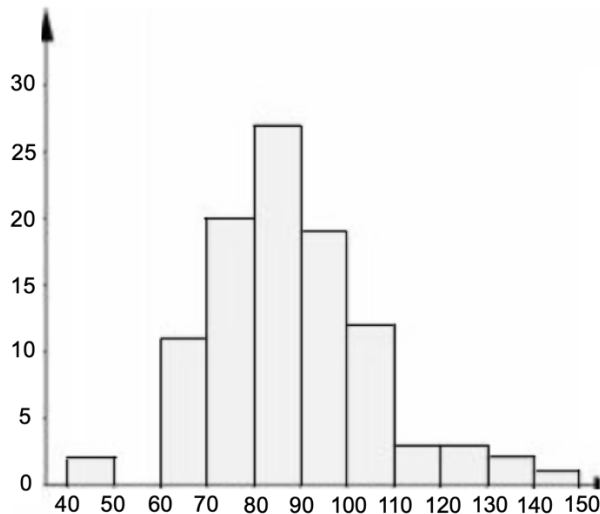
5. Sketch a histogram whose shape is not symmetrical and has more than one peak.

Answers will vary



6. Suppose a histogram of a data set is symmetrical. If an outlier is added to a data set that is far greater than the median, which direction will the graph be skewed?
The graph will be right skewed

Use the histogram for questions 7-10.



7. The shape of the histogram can be described as

- ☐ symmetric.
- ☐ left skewed.
- ☒ right skewed.

8. Is there a gap in the histogram? If so, where?

Yes, between 50 – 60

9. Is there a cluster in the histogram? If so, where?

Yes, between 60 – 110

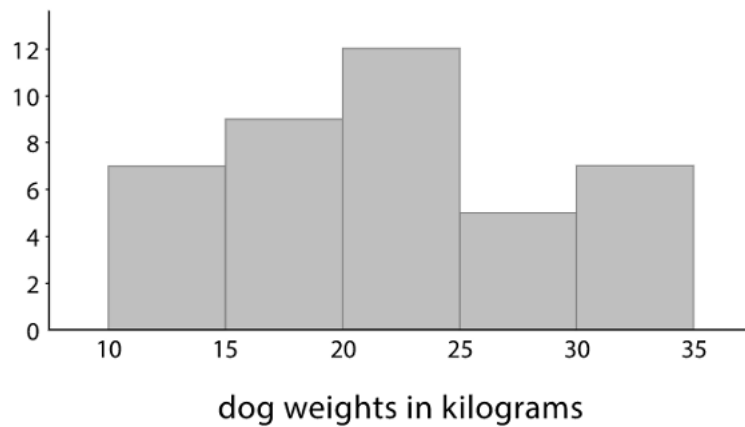
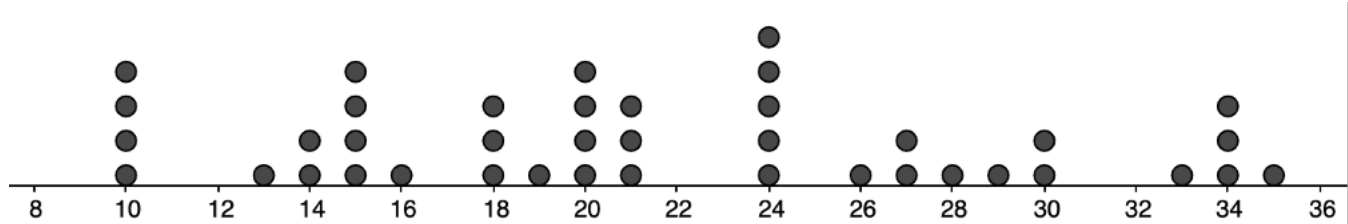
10. If the data value 395 were added to the histogram above, it would be considered an outlier.

Answers will vary

Grade 6 Accelerated
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Lesson 7 Practice

Name **Answer Key** _____
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Here is a histogram and dot plot showing the weights, in pounds, of 40 dogs at a dog show.



- Which data display can be used to determine the mean weight?

Dot plot

- Find the mean weight. Interpret the mean.

$$[4(10) + 13 + 2(14) + 4(15) + 16 + 3(18) + 19 + 4(20) + 3(21) + 5(24) + 26 + 2(27) + 28 + 29 + 2(30) + 33 + 3(34) + 35] \div 40 = 21.5$$

The average is 21.5 kg.

- Which data display can be used to determine the median weight?

Dot plot

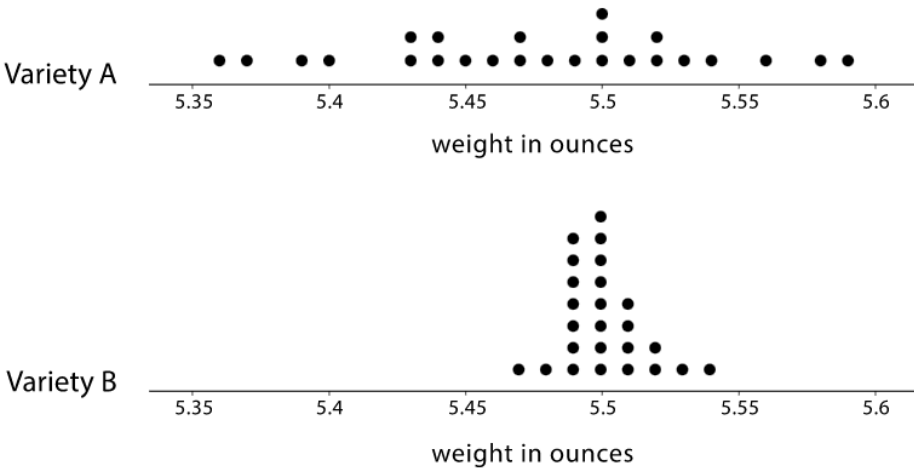
- Find the median weight. Interpret the median.

20.5

The median weight of the dogs is 20.5 kg.

A farmer sells tomatoes in packages of *ten*. She would like the tomatoes in each package to all be about the same size and close to 5.5 ounces in weight.

The farmer is considering two different tomato varieties: Variety *A* and Variety *B*. The dot plots show her data.



- Describe the shape and distribution of Variety *A*.
The data appear symmetric but spread out.
- Describe the shape and distribution of Variety *B*.
The data are clustered and slightly right skewed.
- What percentage of tomatoes in Variety *A* were less than 5.49 ounces?
 $\frac{13}{25} = \frac{52}{100}$ 52%
- What percentage of tomatoes in Variety *B* were less than 5.49 ounces?
 $\frac{2}{25} = \frac{8}{100}$ 8%
- What is the range of weights in Variety *A* and *B*?

Variety <i>A</i>	Variety <i>B</i>
$5.59 - 5.36 = 0.23$	$5.54 - 5.37 = 0.07$

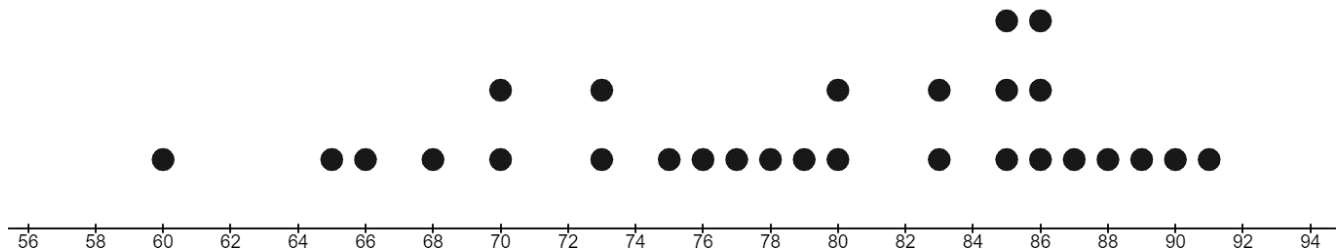
- Which tomato variety should the farmer choose? Explain your reasoning.

The farmer should choose Variety *B* because the tomatoes are all closer to the same size.

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Lesson 8 Practice

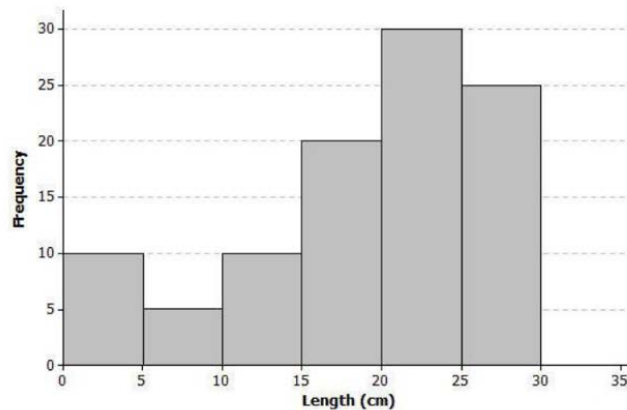
Name Answer Key
Date _____
Period _____

The grades (based on 100 points) of a class of students is shown in the dot plot.



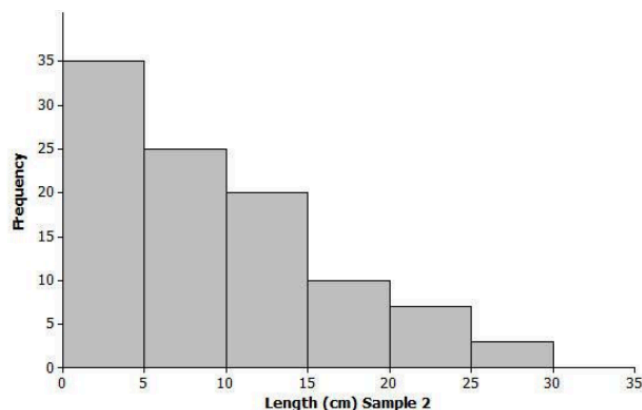
- Describe the spread and shape of the distribution.
The distribution is spread out and slightly left skewed.
- Find and interpret the mean scores. Round to the nearest tenth.
$$[60 + 65 + 66 + 68 + 2(70) + 2(73) + 75 + 76 + 77 + 78 + 79 + 2(80) + 2(83) + 3(85) + 3(86) + 87 + 88 + 89 + 90 + 91] \div 28 = \frac{2214}{28} \approx 79.1$$
The average score of class A is 79.1.
- Find and interpret the median scores.
The median score was 80. Half of the scores were higher, and half of the scores were lower.
- How would the mean and median change if you found the mean and median of the top 10 grades on the test?
Both mean and median would increase.
- Find the mean of the top 10 scores.
$$\frac{2(85) + 3(86) + 87 + 88 + 89 + 90 + 91}{10} = \frac{873}{10} = 87.3$$
- Find the median of the top 10 scores.
86.5

Scientists collected data from many yellow perch because they were concerned about the survival of the yellow perch. Scientists captured yellow perch from a local lake. They recorded data on each fish and then returned each fish to the lake. Consider the following histogram of data on the length (in centimeters) for a sample.



7. Which measure of the center can be determined by the histogram, the mean or the median?
Neither can be exactly determined.
8. Estimate the median length of the yellow perch in the sample.
Answers will vary. Between 20–25 cm.
9. Based on the shape of this data distribution, do you think the mean length of a yellow perch would be greater than, less than, or the same as your estimate of the median?
Less than

Another sample of Great Lake yellow perch from a different lake was collected. A histogram of the lengths for the fish in this sample is shown below.



10. Based on the shape of this data distribution, do you think the median length of a yellow perch from this second sample would be greater than, less than, or the same the median length from the first sample?
The median of the second would be less than the first.

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The number of letters in 20 students' full names is shown below.

6, 7, 8, 8, 10, 11, 12, 12, 12, 12, 13, 15, 15, 16, 18, 19, 19, 21, 24, 32

1. Determine the five-number summary for the data set.

Minimum: 6

First Quartile: 10.5

Median: 12.5

Third Quartile: 18.5

Maximum: 32

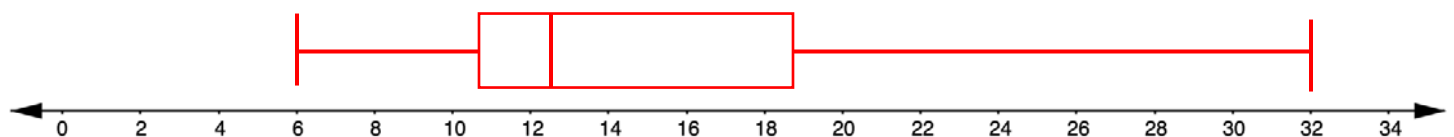
2. What is the range of the data?

$$32 - 6 = 26$$

3. What is the interquartile range of the data?

$$18.5 - 10.5 = 8$$

4. Create a box plot of the number of letters in the students' full names.



A school's data on the number of daily absences for the last 14 days are shown.

7	8	11	6	10	4	13	2	5	8	10	13	6	17
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5. Determine the five-number summary for the data set.

2, 4, 5, 6, 6, 7, 8, 8, 10, 10, 11, 13, 13, 17

Minimum: 2

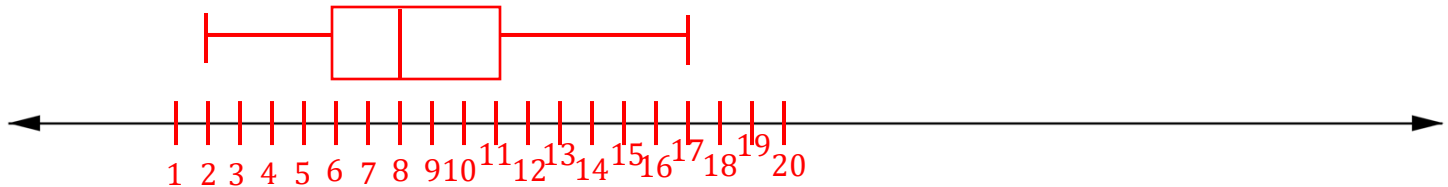
Median: 8

Maximum: 17

First Quartile: 6

Third Quartile: 11

6. Create a box plot of the daily absences.



7. Explain how you determine the scale to use for the number line of your box plot.

Use the range of data to determine an appropriate scale. All of the data values should fit on the number line.

The free-throw percentages of twelve players on a basketball team are shown.

71.2	88.7	50	81.1	84.3	78.7	74.5	82.4	85.1	91.5	76.5	87.2
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8. Determine the five-number summary for the data set.

50, 71.2, 74.5, 76.5, 78.7, 81.1, 82.4, 84.3, 85.1, 87.2, 88.7, 91.5

Minimum: 50

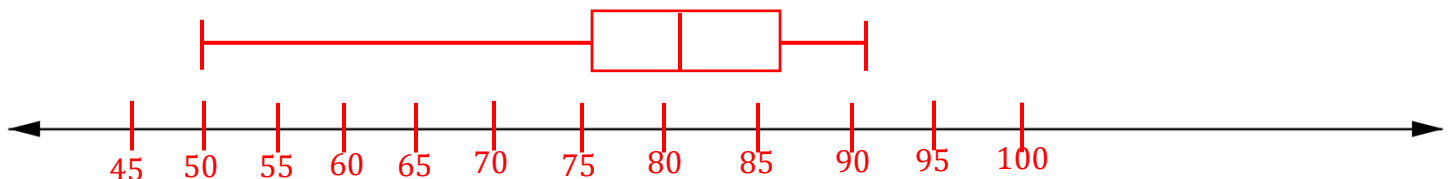
Median: 81.75

Maximum: 91.5

First Quartile: 75.5

Third Quartile: 86.15

9. Create a box plot of the team's free-throw percentages.



10. What is the interquartile range?

$86.15 - 75.5 = 10.65$

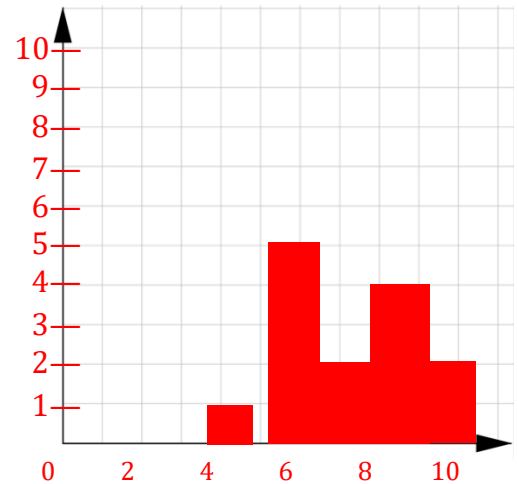
Mr. Gonzalez is tracking the number of hours he sleeps each night. His data for the last 2 weeks is shown.

6.5	8	5.5	7.5	9	10	6	7	5.5	8	6.5	10	4	8.5
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1. Create a frequency chart of the data. 2. Create a histogram of the data.

Hours of Sleep	Frequency
4 – 5.2	1
5.3 – 6.5	5
6.6 – 7.8	2
7.9 – 9.1	4
9.2 – 10.4	2

Range: $10 - 4 = \frac{6}{5 \text{ intervals}} = 1.2$

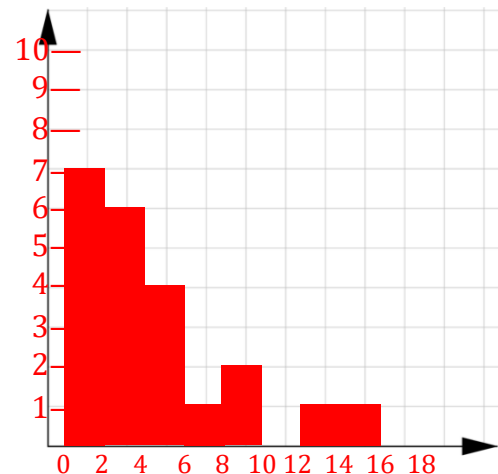


Sole asked her classmates how many pairs of shoes they have. Her data are shown.

1, 1, 1 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 5, 5, 5, 6, 8, 9, 10, 13, 15

3. Create a frequency chart of the data. 4. Create a histogram of the data.

Pairs of Shoes	Frequency
1 – 2	7
3 – 4	6
5 – 6	4
7 – 8	1
9 – 10	2
11 – 12	0
13 – 14	1
15 – 16	1



5. Explain how you determined the scale for the axes on the histogram in question 4.
Answers may vary. I used the interval width, 2, to number the x-axis: $16 \div 8 = 2$.
Because the frequency does not exceed 7, I numbered the y-axis by 1.

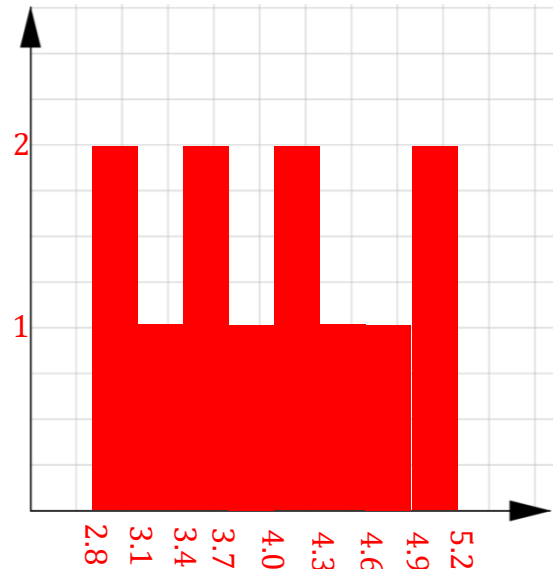
The weights, in pounds (lb), of 12 Dachshund puppies are shown.

2.8	3.6	3.1	5.2	4.8	3.9	4	4.3	5	3.5	2.9	4
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6. Create a frequency chart of the data.

Weight (lb)	Frequency
2.8 – 3.1	2
3.1 – 3.4	1
3.4 – 3.7	2
3.7 – 4.0	1
4.0 – 4.3	2
4.3 – 4.6	1
4.6 – 4.9	1
4.9 – 5.2	2

7. Create a histogram of the data.



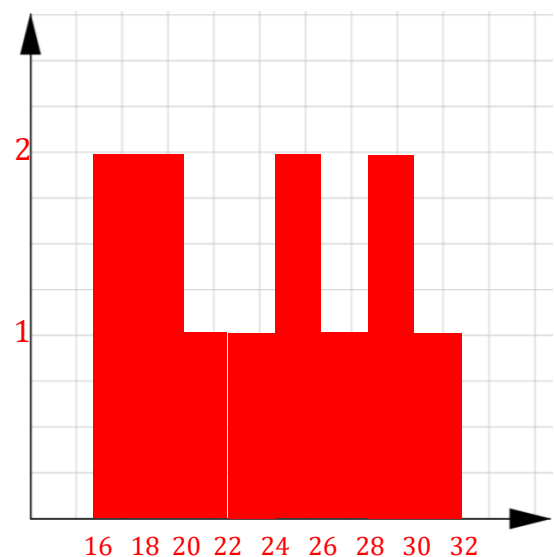
The weights, in pounds (lb), of 12 adult Dachshunds are shown.

25	18	32	16	27	30	26	19	29	20	22	24
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8. Create a frequency chart of the data.

Weight (lb)	Frequency
16 – 18	2
18 – 20	2
20 – 22	1
22 – 24	1
24 – 26	2
26 – 28	1
28 – 30	2
30 – 32	1

9. Create a histogram of the data.



10. How are your histograms for questions 7 and 9 different?

Answers may vary. Because the ranges are different, the intervals for the table and x -axis are different. The y -axis intervals are the same because the frequencies (1 or 2) were the same, so the graphs look similar but are different. The histogram for 7 has one more peak than 9 does.